Listing 11.14 shows an implementation of HKDF-Expand using HMAC-SHA-256. To generate the required amount of output key material, HKDF-Expand performs a loop. Each iteration of the loop runs HMAC to produce a block of output key material with the following inputs:

1. The HMAC tag from the last time through the loop, unless this is the first loop.
2. The context string.
3. A block counter byte, which starts at 1 and is incremented each time.

With HMAC-SHA-256 each iteration of the loop generates 32 bytes of output key material, so you'll typically only need one or two loops to generate a big enough key for most algorithms. Because the block counter is a single byte, and cannot be 0, you can only loop a maximum of 255 times, which gives a maximum key size of 8,160 bytes. Finally, the output key material is converted into a `Key` object using the `javax.crypto.spec.SecretKeySpec` class. Create a new file named HKDF.java in the src/main/java/com/manning/apisecurityinaction folder with the contents of the file.

> **TIP** If the master key lives in a HSM or KMS then it is much more efficient to combine the inputs into a single byte array rather than making multiple calls to the `update()` method.

Listing 11.14 HKDF-Expand

```
package com.manning.apisecurityinaction;
```